EFFECTS OF SOCIOECONOMIC FACTORS ON CHILDREN MORTALITY RATE IN VIETNAM.

My Le’25

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MOTIVATION

- Being an IVF child:
  - A deep connection with those who are "mothers."
  - Children's health issues and their socioeconomic factors.
- Background in Quantitative Economics
  - Deepen my understanding of the socioeconomic factors that contribute to children mortality rate.
  - Specifically in the context of Vietnam.

DATA COLLECTION

Vietnam Household Living Standards Survey 2020

- Average healthcare expenditure per person having treatment in the past 12 months by region in VND

Vietnam Statistical Yearbook 2016 & 2022

- Infant mortality rate by Region in percentage
- Under-five mortality rate by Region in percentage
- Monthly average income per capita at current prices by region in VND
- Percentage of population using hygienic water source

HYPOTHESIS

Ho: Socioeconomic factors have no significant effect on children mortality rate in Vietnam

Ha: Socioeconomic factors have some significant effects on children mortality rate in Vietnam

EMPIRICAL MODELS

01 $$Infant\ mortality\ rate = \beta_0 + \beta_1 Income + \beta_2 Healthcare\ Expenditure + \beta_3 Hygienic\ Water + \mu$$

02 $$Under\ five\ mortality\ rate = \beta_0 + \beta_1 Income + \beta_2 Healthcare\ Expenditure + \beta_3 Hygienic\ Water + \mu$$

INFANT MORTALITY RATE AND ITS FACTORS

```
. regress InfantMortalityRateinPercent Monthlyincpercapita Averagehealthcareexppercapit Percentagehygienicwatersource
```

Source	SS	df	MS
Model	.057796527	3	.019265509
Residual	.062479779	32	.001952493
Total	.120276306	35	.003436466

Number of obs	=	36
F(3, 32)	=	9.87
Prob > F	=	0.0001
R-squared	=	0.4805
Adj R-squared	=	0.4318
Root MSE	=	.04419

InfantMortalityRateinPercent	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
Monthlyincpercapita	-2.79e-08	9.52e-09	-2.93	0.006	-4.73e-08	-8.49e-09
Averagehealthcareexppercapit	2.24e-08	1.09e-08	2.07	0.047	3.28e-10	4.46e-08
Percentagehygienicwatersource	-.1786054	.1336298	-1.34	0.191	-.4508005	.0935896
_cons	.3488681	.1143364	3.05	0.005	.1159724	.5817638

UNDER FIVE MORTALITY RATE AND ITS FACTORS

```
. regress UnderfivemortalityratebyPer Monthlyincpercapita Averagehealthcareexppercapit Percentagehygienicwatersource

      Source |       SS           df       MS      Number of obs   =        36
-------------+----------------------------------   F(3, 32)        =      5.45
       Model |  .099369684         3  .033123228   Prob > F        =    0.0038
    Residual |  .194565205        32  .006080163   R-squared       =    0.3381
-------------+----------------------------------   Adj R-squared   =    0.2760
       Total |  .293934889        35   .00839814   Root MSE        =    .07798

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  UnderfivemortalityratebyPer |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
------------------------------+-----------------------------------------------------------------
          Monthlyincpercapita |  -5.94e-09   1.68e-08    -0.35   0.726    -4.02e-08    2.83e-08
 Averagehealthcareexppercapit |  -1.01e-08   1.92e-08    -0.53   0.603    -4.91e-08    2.90e-08
Percentagehygienicwatersource |  -.5741113   .2358122    -2.43   0.021    -1.054445   -.0937776
                        _cons |   .8072978   .2017658     4.00   0.000     .3963144    1.218281
------------------------------------------------------------------------------------------------
```

CORRELATION COEFFICIENTS

	Monthly income	Healthcare Expenditure	Hygienic Water Source
Infant Mortality Rate	-2.79	**2.24**	-0.17
Under-five Mortality Rate	-5.94	**-1.01**	-0.57

RESULTS

- **Two factors** in the models share the same trend:
- But for **Healthcare Expenditure**:
  - **Positive correlation:** a 1000 VND increase in average healthcare spending leads to a 2.24% increase in *Infant mortality rate.*
  - **Negative correlation:** a 1000 VND increase in average healthcare spending leads to a 1.01% decrease in the *Under-five mortality rate.*
- Why?
  - The **leading causes** of mortality among infants and older children:
    - Infants are more vulnerable to perinatal conditions, congenital anomalies, and preterm birth complications.
    - Older children may face more infectious diseases and injuries.
- Healthcare spending may be more effective in addressing the latter, leading to a greater reduction in mortality rates.

STATISTICALLY SIGNIFICANT?

Statitical significance	Monthly income	Healthcare Expenditure	Hygienic Water Source
Infant Mortality Rate	0.006	0.047	**0.191**
Under-five Mortality Rate	0.726	0.603	**0.021**
Under-five Mortality Rate after Robust	0.786	0.646	**0.008**

RESULTS

- For Infant Mortality Rate:
  - Monthly income and Healthcare expenditure are statically significant at a 5% level.
  - Hygienic water source is not a significant factor.
- For Under-five Mortality Rate:
  - Monthly income and Healthcare expenditure are not a significant factor.
  - Hygienic water source is statistically significant at 1% level.
- How can this **reverse trend** happen?
  - **Newborns and babies up to 6 months old should never drink water. They only need to drink breastmilk or infant formula*.**

Source: Hackensack Meridian Health

LIMITATIONS

- Effect of COVID-19
- Omitted Variables Bias
- Specific Cultural Context

NEXT STEPS

- Comparison among different regions
- More profound research into the causes of previously mentioned reverse tendencies.